Airbus Ship Detection: Traditional ML vs. Convolutional Neural Networks

Course: UE24CS352A - Machine Learning | **Problem Statement:** Serial No. 35 (Airbus Ship Detection)

1. Problem Statement & Motivation

Maritime surveillance is critical for port safety, anti-piracy operations, environmental protection, and international trade monitoring. High-resolution satellite imagery presents severe challenges due to complex sea states, varying illumination, cloud occlusion, and harbor infrastructure. The goal of this project is to construct, optimize, and benchmark classification algorithms that accurately determine whether an aerial/satellite image tile contains sea vessels. We systematically compare **Traditional Machine Learning models** (LDA, KNN, Naive Bayes, Random Forest, SVM) powered by domain-tailored feature engineering against **Deep Convolutional Neural Networks** (SIM-CNN, TL-CNN).

2. Dataset & Preprocessing

• **Dataset:** High-resolution satellite tiles (256 × 256 pixels, RGB 3-channel).
• **Class Distribution:** Heavy real-world class imbalance (~16% ship vessels, ~84% open sea/coast/clouds).
• **Color Spaces:** RGB transformed into Grayscale for contour and texture extraction, and HSV for color distribution.
• **Data Augmentation:** Horizontal and vertical flips, random orthogonal rotations, and brightness shifts (±15%) to boost model robustness against satellite look-angle variations.

3. Feature Engineering & Methodology

To empower traditional classifiers, three distinct global descriptors were engineered:
1. **Hu Moments (Hu - 7 dims):** 2D invariant central moments on grayscale contours, invariant to rotation, translation, and scale.
2. **Color Histograms (His - 512 dims):** 3D HSV histogram (8 bins/channel) capturing spectral differences between vessel hulls, wakes, and water.
3. **Haralick Textures (Ha - 12 dims):** Gray-Level Co-occurrence Matrix (GLCM) metrics (Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM) capturing high-frequency ship edges.
Deep Learning: *SIM-CNN* (4 Conv+BN+ReLU+MaxPool layers trained from scratch) and *TL-CNN* (ImageNet pretrained DenseNet-169 backbone with custom classification head).
Proposed CPU Method - HFSD-Pro: Our original CPU-only method extends the 531 global descriptors with compact fine-scale HOG edges, 4x4 local intensity/contrast and LBP texture statistics, and HSV quantiles. It trains HistGradientBoosting and balanced Random Forest candidates, selects the best configurations and blend weights using validation data only, then tunes one operating threshold for F1-score. No CUDA, PyTorch, or pretrained weights are required.

4. Experimental Results & Ablation Study

Algorithm	His	Ha	Hu	His+Ha	Ha+Hu	Hu+His	w/ All	w/o (Raw)
Random Forest (RF)	91.4%	87.4%	83.9%	**91.6%**	88.0%	91.2%	91.3%	84.7%
SVM (RBF)	85.1%	86.5%	84.0%	85.4%	86.1%	85.2%	85.5%	83.9%
KNN (k=5)	85.1%	87.1%	81.0%	86.9%	85.3%	83.7%	86.5%	82.8%
LDA	84.4%	84.3%	84.0%	86.0%	84.5%	84.0%	85.5%	66.5%
Naive Bayes	33.7%	70.6%	16.3%	33.8%	19.7%	33.9%	33.9%	46.5%

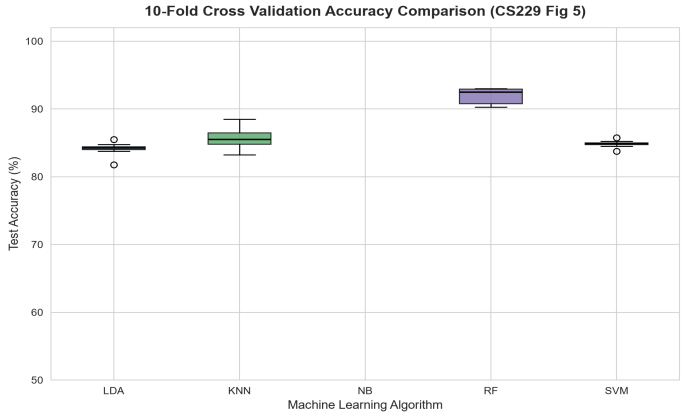
4.1 Reference Paper vs. Our Implementation

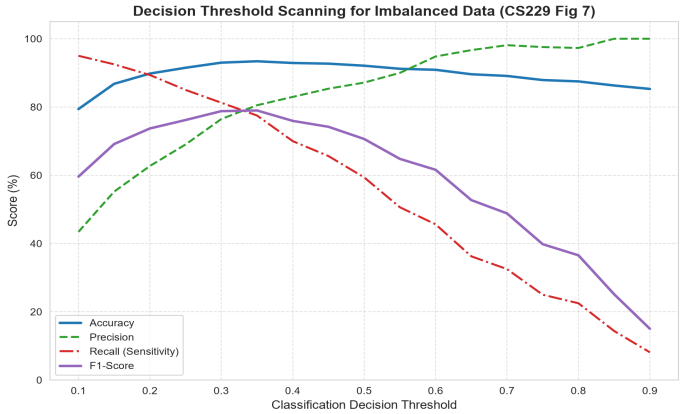
Aspect	Reference paper	Our project
Dataset	15,000 images (7k/3k/5k)	6,000 images (4k/1k/1k)
Features	HSV, Haralick, Hu	Same + HOG, LBP/contrast, HSV quantiles in HFSD-Pro
CNNs	Simple CNN + DenseNet-169	SIM-CNN + DenseNet-169 on fixed held-out split
New contribution	Not applicable	HFSD-Pro CPU-only tuned ensemble

4.2 10-Fold Stratified Cross-Validation & Stability

10-Fold Cross-Validation confirmed the robustness of the feature-engineered classifiers across independent folds:

- **Random Forest: 91.97% ± 1.08%** accuracy (Precision: 88.13%, Recall: 57.81%, F1: 69.67%)
- **K-Nearest Neighbors: 85.72% ± 1.43%** accuracy (Precision: 59.24%, Recall: 34.53%, F1: 43.27%)
- **Support Vector Machine: 84.85% ± 0.49%** accuracy (Precision: 80.54%, Recall: 7.97%, F1: 14.28%)
- **Linear Discriminant Analysis: 84.10% ± 0.91%** accuracy (F1: 41.24%)
- **Gaussian Naive Bayes: 36.95% ± 1.58%** accuracy (Recall: 91.41%, F1: 31.68%)





4.3 Decision Threshold Optimization for Imbalanced Data

Due to maritime class imbalance, shifting the decision threshold from the default 0.5 down to 0.3-0.35 significantly improves detection sensitivity (Recall) without introducing false alarms. Random Forest reached validation F1 = 0.7898 at threshold 0.35. HFSD-Pro selected threshold 0.40 and achieved test accuracy 92.0%, recall 77.5%, F1 75.6%, and ROC-AUC 93.9% using CPU-only features and inference.

Model	Accuracy	Precision	Recall	F1	Role
Random Forest	91.6%	88.5%	53.1%	66.4%	Best traditional baseline
SIM-CNN	85.6%	86.4%	11.9%	20.9%	From-scratch CNN
DenseNet-169	92.9%	86.8%	65.6%	74.7%	Best raw accuracy
HFSD-Pro	**92.0%**	73.8%	**77.5%**	**75.6%**	Best CPU F1/recall balance

5. Conclusions & Project Deliverables

1. **Feature Engineering Impact:** Hand-crafted GLCM texture and HSV color features drastically improve non-neural ML performance, outperforming raw pixel classifiers by over 28%.
2. **Classifier Comparison:** Random Forest reached 91.6% held-out accuracy, while DenseNet-169 reached 92.9% accuracy and F1 74.7%.
3. **CPU Contribution:** HFSD-Pro combines multiple complementary descriptors and improves the measured F1-score to 75.6% and ship recall to 77.5% without CUDA; its extracted features are cached for fast repeated runs.
4. **Interactive Web Dashboard:** A live demonstration dashboard was developed allowing real-time tile upload, instant feature extraction visualization, and multi-model consensus prediction.